In [ ]:
```
import numpy as np
```

In [ ]:
```
!pip install kaggle
```

```
Requirement already satisfied: kaggle in /usr/local/lib/python3.12/dist-packages
(2.0.0)
Requirement already satisfied: bleach in /usr/local/lib/python3.12/dist-packages
(from kaggle) (6.3.0)
Requirement already satisfied: kagglesdk<1.0,>=0.1.15 in /usr/local/lib/python3.1
2/dist-packages (from kaggle) (0.1.16)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packag
es (from kaggle) (26.0)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-package
s (from kaggle) (5.29.6)
Requirement already satisfied: python-dateutil in /usr/local/lib/python3.12/dist-
packages (from kaggle) (2.9.0.post0)
Requirement already satisfied: python-slugify in /usr/local/lib/python3.12/dist-p
ackages (from kaggle) (8.0.4)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-package
s (from kaggle) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (f
rom kaggle) (4.67.3)
Requirement already satisfied: urllib3>=1.15.1 in /usr/local/lib/python3.12/dist-
packages (from kaggle) (2.5.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-pac
kages (from bleach->kaggle) (0.5.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-package
s (from python-dateutil->kaggle) (1.17.0)
Requirement already satisfied: text-unidecode>=1.3 in /usr/local/lib/python3.12/d
ist-packages (from python-slugify->kaggle) (1.3)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python
3.12/dist-packages (from requests->kaggle) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-pac
kages (from requests->kaggle) (3.11)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/di
st-packages (from requests->kaggle) (2026.2.25)
```

In [ ]:
```
from google.colab import files
files.upload()
```

In [ ]:
```
!ls
```

```
'Global_Superstore2 (1).csv'  'Global_Superstore2 .csv'
 Global_Superstore2.csv        sample_data
```

In [ ]:
```
!unzip global-super-store-dataset.zip
```

```
unzip:  cannot find or open global-super-store-dataset.zip, global-super-store-da
taset.zip.zip or global-super-store-dataset.zip.ZIP.
```

In [ ]:
```
!ls
```

```
'Global_Superstore2 (1).csv'  'Global_Superstore2 .csv'
 Global_Superstore2.csv        sample_data
```

In [ ]:
```
import zipfile
```

In [ ]:
```
import os
```

In [ ]:

In [ ]:
```
import pandas as pd

try:
    df = pd.read_csv('/content/Global_Superstore2.csv', encoding='latin1')
except UnicodeDecodeError:
    df = pd.read_csv('/content/Global_Superstore2.csv', encoding='ISO-8859-1')
```

In [ ]:
```
df.head()
```

Out[ ]:

	ï»¿Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segme
0	40098	CA-2014-AB10015140-41954	11/11/2014	11/13/2014	First Class	AB-100151402	Aaron Bergman	Consun
1	26341	IN-2014-JR162107-41675	2/5/2014	2/7/2014	Second Class	JR-162107	Justin Ritter	Corpor
2	25330	IN-2014-CR127307-41929	10/17/2014	10/18/2014	First Class	CR-127307	Craig Reiter	Consun
3	13524	ES-2014-KM1637548-41667	1/28/2014	1/30/2014	First Class	KM-1637548	Katherine Murray	Ho Off
4	47221	SG-2014-RH9495111-41948	11/5/2014	11/6/2014	Same Day	RH-9495111	Rick Hansen	Consun

5 rows × 24 columns

In [ ]:
```
data = df.copy()
data
```

Out[ ]:

	ï»¿Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
0	40098	CA-2014-AB10015140-41954	11/11/2014	11/13/2014	First Class	AB-100151402	Aaron Bergman
1	26341	IN-2014-JR162107-41675	2/5/2014	2/7/2014	Second Class	JR-162107	Justin Ritter
2	25330	IN-2014-CR127307-41929	10/17/2014	10/18/2014	First Class	CR-127307	Craig Reiter
3	13524	ES-2014-KM1637548-41667	1/28/2014	1/30/2014	First Class	KM-1637548	Katherine Murray
4	47221	SG-2014-RH9495111-41948	11/5/2014	11/6/2014	Same Day	RH-9495111	Rick Hansen
...	...	...	...	...	...	...	...
995	24278	IN-2015-JF1541527-42095	4/1/2015	4/5/2015	Standard Class	JF-1541527	Jennifer Ferguson
996	36249	CA-2015-AS10225140-42347	12/9/2015	12/13/2015	Second Class	AS-102251404	Alan Schoenberger
997	30517	IN-2014-CK1259592-41929	10/17/2014	10/19/2014	Second Class	CK-1259592	Clytie Kelty
998	1408	MX-2014-SS2014018-41658	1/19/2014	1/26/2014	Standard Class	SS-2014018	Saphhira Shifley
999	20944	ID-2015-RP19855130-42348	12/10/2015	12/14/2015	Standard Class	RP-19855130	Roy Phan

1000 rows × 24 columns

In [ ]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 24 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   ï»¿Row ID       1000 non-null   int64
 1   Order ID        1000 non-null   object
 2   Order Date      1000 non-null   object
 3   Ship Date       1000 non-null   object
 4   Ship Mode       1000 non-null   object
 5   Customer ID     1000 non-null   object
 6   Customer Name   1000 non-null   object
 7   Segment         1000 non-null   object
 8   Postal Code     194 non-null    float64
 9   City            1000 non-null   object
 10  State           1000 non-null   object
 11  Country         1000 non-null   object
 12  Region          1000 non-null   object
 13  Market          1000 non-null   object
 14  Product ID      1000 non-null   object
 15  Category        1000 non-null   object
 16  Sub-Category    1000 non-null   object
 17  Product Name    1000 non-null   object
 18  Sales           1000 non-null   float64
 19  Quantity        1000 non-null   int64
 20  Discount        1000 non-null   float64
 21  Profit          1000 non-null   float64
 22  Shipping Cost   1000 non-null   float64
 23  Order Priority  1000 non-null   object
dtypes: float64(5), int64(2), object(17)
memory usage: 187.6+ KB
```

In [ ]:
```
data['Market'].value_counts()
```

Out[ ]:

Market	count
Asia Pacific	365
Europe	248
USCA	200
LATAM	133
Africa	54

dtype: int64

In [ ]:
```
data['City'].value_counts()
```

Out[ ]:

City	count
New York City	25
Los Angeles	11
San Francisco	10
Gold Coast	10
London	10
...	...
Barletta	1
Angers	1
Kumasi	1
CancÃºn	1
Kitwe	1

592 rows × 1 columns

dtype: int64

In [ ]:
```
data['Country'].value_counts()
```

Out[ ]:

Country	count
United States	194
Australia	85
France	76
China	75
India	64
...	...
Benin	1
Bolivia	1
Chile	1
Martinique	1
Syria	1

93 rows × 1 columns

dtype: int64

In [ ]:
```
data['Region'].value_counts()
```

Out[ ]:

Region	count
Western Europe	131
Oceania	100
Eastern Asia	90
Southern Asia	87
Central America	77
Southeastern Asia	70
Western US	62
Eastern US	60
Northern Europe	48
Central US	46
Southern Europe	46
South America	37
Southern US	26
Eastern Europe	23
Caribbean	19
Western Asia	18
Central Africa	14
North Africa	12
Southern Africa	11
Eastern Africa	9
Western Africa	8
Canada	6

dtype: int64

In [ ]:
```
data.isnull().sum()
```

Out[ ]:

	0
ï»¿Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Postal Code	806
City	0
State	0
Country	0
Region	0
Market	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0
Shipping Cost	0
Order Priority	0

dtype: int64

In [ ]:
```
data['Postal Code'] =data['Postal Code'].fillna('unknown')
```

In [ ]:
```
data.isnull().sum()
```

Out[ ]:

	0
ï»¿Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Postal Code	0
City	0
State	0
Country	0
Region	0
Market	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0
Shipping Cost	0
Order Priority	0

dtype: int64

In [ ]:

```
data.describe()
```

Out[ ]:

	ï»¿Row ID	Sales	Quantity	Discount	Profit	Shipping Cost
count	1000.000000	1000.000000	1000.00000	1000.000000	1000.000000	1000.000000
mean	25079.328000	1710.971470	5.55800	0.092840	288.920440	272.384897
std	12897.726632	1259.239238	2.71846	0.148666	574.504782	176.160716
min	58.000000	1.910000	1.00000	0.000000	-3059.820000	1.070000
25%	15118.750000	826.907500	4.00000	0.000000	10.037500	209.827500
50%	25084.500000	1585.115000	5.00000	0.000000	190.685000	258.897500
75%	34524.000000	2477.812500	7.00000	0.150000	518.872500	351.070250
max	51284.000000	9892.740000	14.00000	0.800000	4946.370000	923.630000

In [ ]:

```
data.nunique()
```

Out[ ]:

	0
ï»¿Row ID	1000
Order ID	867
Order Date	599
Ship Date	613
Ship Mode	4
Customer ID	800
Customer Name	496
Segment	3
Postal Code	77
City	592
State	341
Country	93
Region	22
Market	5
Product ID	529
Category	3
Sub-Category	17
Product Name	529
Sales	946
Quantity	14
Discount	20
Profit	956
Shipping Cost	971
Order Priority	4

dtype: int64

In [ ]:

```
data['Ship Mode'].value_counts()
```

Out[ ]:

Ship Mode	count
Standard Class	343
First Class	317
Second Class	239
Same Day	101

dtype: int64

In [ ]:
```
data['Category'].value_counts()
```

Out[ ]:

Category	count
Technology	395
Furniture	326
Office Supplies	279

dtype: int64

In [ ]:
```
data['Order Date'] = pd.to_datetime(data['Order Date'])
data['Ship Date'] = pd.to_datetime(data['Ship Date'])
```

In [ ]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 24 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   ï»¿Row ID       1000 non-null   int64
 1   Order ID        1000 non-null   object
 2   Order Date      1000 non-null   datetime64[ns]
 3   Ship Date       1000 non-null   datetime64[ns]
 4   Ship Mode       1000 non-null   object
 5   Customer ID     1000 non-null   object
 6   Customer Name   1000 non-null   object
 7   Segment         1000 non-null   object
 8   Postal Code     1000 non-null   object
 9   City            1000 non-null   object
 10  State           1000 non-null   object
 11  Country         1000 non-null   object
 12  Region          1000 non-null   object
 13  Market          1000 non-null   object
 14  Product ID      1000 non-null   object
 15  Category        1000 non-null   object
 16  Sub-Category    1000 non-null   object
 17  Product Name    1000 non-null   object
 18  Sales           1000 non-null   float64
 19  Quantity        1000 non-null   int64
 20  Discount        1000 non-null   float64
 21  Profit          1000 non-null   float64
 22  Shipping Cost   1000 non-null   float64
 23  Order Priority  1000 non-null   object
dtypes: datetime64[ns](2), float64(4), int64(2), object(16)
memory usage: 187.6+ KB
```

In [ ]: